, .

$$h = \frac{\cos\alpha \pm \sqrt{\cos^2\alpha - 4\frac{h_0}{R}\sin\alpha}}{2\frac{\sin\alpha}{R}} \quad \Rightarrow h_1 = 16,5mm,\ h_2 = 83,5mm$$

$H < h_2$ h_1 . ,

h_2 .

3.5 **[1.0점]** H , (12) .

$$h_0 > \frac{R}{4tg\alpha}\cos\alpha\,, \qquad (14)$$

$$\sin\alpha > \frac{R}{4h_0} = 0,018\,.$$

4. 물의 유출 (3.0점)

4.1 **[3.0점]** .

.

$$2\sigma_0\pi h^2 < 2\frac{\pi R^2 h^2}{6}\rho g\,, \qquad (15)$$

$$R > \sqrt{\frac{6\sigma_0}{\rho g}} = 6,6mm \qquad (16)$$

채점 기준

№	평가 항목	배점	
1		0,5	**1,0**
	(2) $(\sigma_2 - \sigma_1)\Delta l = \sigma_0 \Delta l \cos\theta$	0,5	
2.1	(4) $\Delta U_S = -\sigma_0 \cos\theta \cdot \Delta S = -\sigma_0 \cos\theta \cdot 2\pi R\Delta h$	0.5	**0.5**
2.2	(5) $\Delta U_G = \pi R^2 \Delta h \rho g h$	0,5	**0,5**
2.3	(6) $\sigma_0 \cos\theta \cdot 2\pi R\Delta h = \pi R^2 \Delta h \rho g h \quad \Rightarrow \quad h_0 = \frac{2\sigma_0\cos\theta}{\rho g R}$	0,5	**1,0**
	(7) $h_0 = \frac{2\sigma_0\cos\theta}{\rho g R}$	0,2	
	( ) $h_0 = 1,4\cdot 10^{-2} m = 14mm$	0.3	
3.1	(8) $\Delta U_S = -\sigma_0 \cos\theta \cdot \Delta S = -\sigma_0 \cos\theta \cdot 2\pi r \frac{\Delta h}{\cos\alpha}$	0,5	**0,5**
3.2	(9) $\Delta U_G = \pi r^2 \Delta h \rho g h$	0,5	**0,5**
3.3	(10) $\sigma_0 \cos\theta \cdot 2\pi r \frac{\Delta h}{\cos\alpha} = \pi r^2 \Delta h \rho g h$	0,5	**1.0**